SUPPORT PREPARATION

Shared Support Set S

K real + K fake

Real c = 0



Fake c = 1

QUERY INFERENCE

Shared Query Set Q



One query at a time

Query-Conditioned Prototype Mixing

Image Encoder

CLIP · block 12

K K

μ_0 μ_1

1024-D

Support Feature Bank

z_{ci}

Cached source state

No task-learned parameters

μ_c

z_{ci} μ_c

Query-dependent readout

Image Encoder

CLIP · block 12

q

Per-class softmax

β = 15

0.5 $c \in \{0, 1\}$

0.5

$m_c(q)$

$\tilde{\mu}_c(q)$

$s_p = 15\, q^T(\tilde{\mu}_1(q) - \tilde{\mu}_0(q))$

Similarity margin

Stage-wise Visual Prompting

y

Prompted CLIP

$\mathcal{L}CE$

t_0, t_1

Fixed text anchors

"real" "fake"

Only prompts updated

Fitted prompt parameters

4 × 2 × 1024 = 8,192

Frozen backbone · fitted prompts

2 prompts / stage · replace at 1 / 7 / 13 / 19

1–6 7–12 13–18 19–24

768-D

Fixed text anchors

"real" "fake"

t_0, t_1

Similarity

$p_v = \sigma(15\, v^T(t_1 - t_0))$

8,192 fitted prompt parameters

$p_p = \sigma(s_p)$

0.5 0.5

$p = (p_p + p_v)/2$

Real / Fake

p > 0.5: fake

Real Fake Frozen backbone / text Trainable prompt Image / feature flow Fitted prompt dependency